Speed, Distance and Time — Paper C

HARD · 50 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formula and no conversion factor are given on this paper.
- Where a question asks for an average speed, it means total distance divided by total time.

SECTION 1 — JOURNEYS IN PARTS

1. A car travels 60 km at 60 km/h, then a further 60 km at 120 km/h.

(a) Calculate the time taken for each part of the journey. [2]

(b) Calculate the average speed for the whole journey. [2]

(c) Explain why the answer is not 90 km/h. [1]

2. An athlete runs 500 m in 50 s, then walks a further 500 m in 200 s.

(a) Calculate the total distance and the total time. [2]

(b) Calculate the average speed for the whole 1000 m, in m/s. [2]

(c) State why the answer is not the average of 10 m/s and 2.5 m/s. [1]

SECTION 2 — USING SPEED IN THE REAL WORLD

3. During a storm a student sees a flash of lightning and hears the thunder 6 seconds later. Sound travels at 330 m/s.

(a) Calculate how far away the lightning was, in metres. [2]

(b) Give your answer in kilometres. [1]

(c) Explain why it is reasonable to assume the light arrived instantly. [1]

4. Put these three speeds in order, slowest first. You must show your working. [4]

`15 m/s     50 km/h     840 m/min`

SECTION 3 — WORKING BACKWARDS

5. A train must cover 300 km in 4 hours. It travels the first 100 km at an average speed of 50 km/h.

(a) Calculate the total time allowed for the journey, and the time used so far. [2]

(b) Calculate the distance still to travel and the time remaining. [1]

(c) Calculate the average speed needed for the rest of the journey. [2]

6. Explain, in terms of time, why you can never find an average speed by averaging the two speeds of a two-part journey. [3]

7. A journey is recorded below. [4]

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Time (min):     0     5    10    15    20
Distance (km):  0     3     6     6    12
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Calculate the speed of the fastest section of this journey, in km/h.

End of paper. Check every answer has a unit or a form the question actually asked for.